ISEL
INSTITUTO SUPERIOR DE ENGENHARIA DE LISBOA

Artificial Intelligence course

6th Semester

Bachelor in Informatics and Computer Engineering

Optimization algorithms

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Copyright note

- These slides were partially based on the book:
  El-Ghazali Talbi, *Metaheuristics – From Design to Implementation*, 2009, Wiley

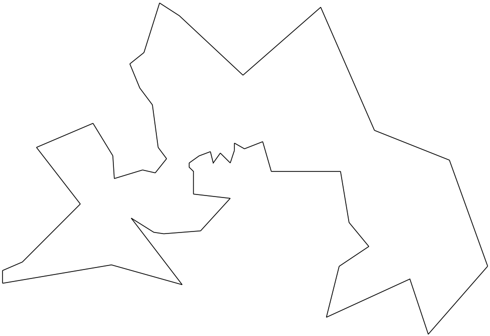

FIGURE 1.4 TSP instance with 52 cities.

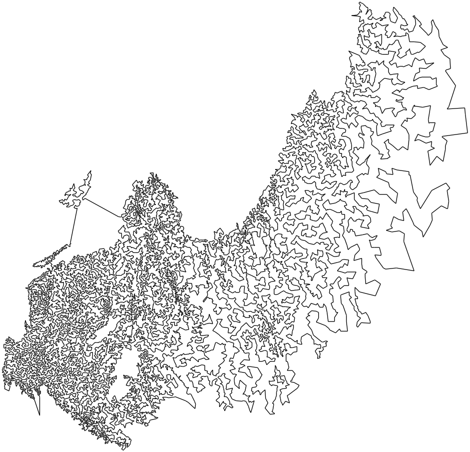

FIGURE 1.5 TSP instance with 24,978 cities.

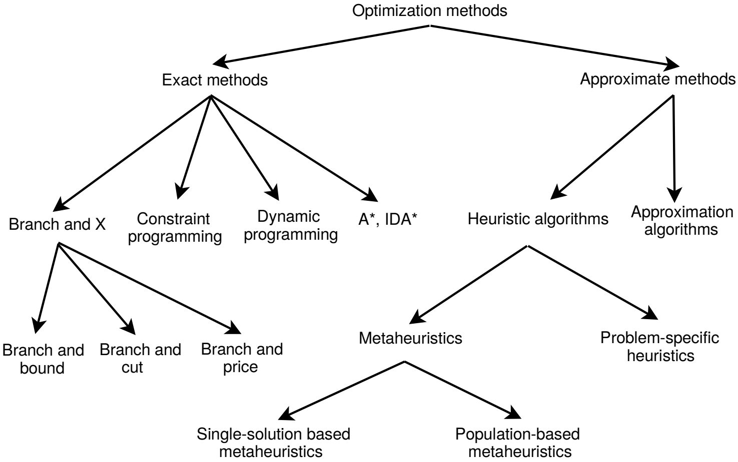

FIGURE 1.7 Classical optimization methods.

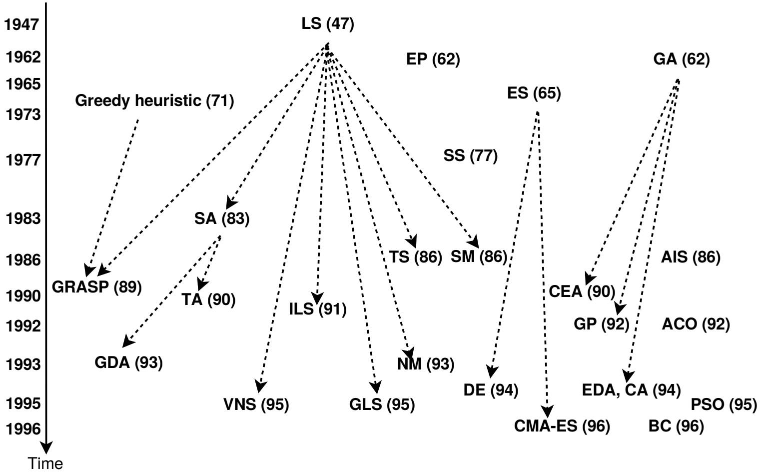

FIGURE 1.8 Genealogy of metaheuristics. The application to optimization and/or machine learning is taken into account as the original date.

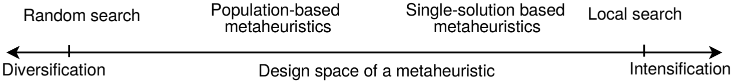

FIGURE 1.9 Two conflicting criteria in designing a metaheuristic: exploration (diversification) versus exploitation (intensification). In general, basic single-solution based metaheuristics are more exploitation oriented whereas basic population-based metaheuristics are more exploration oriented.

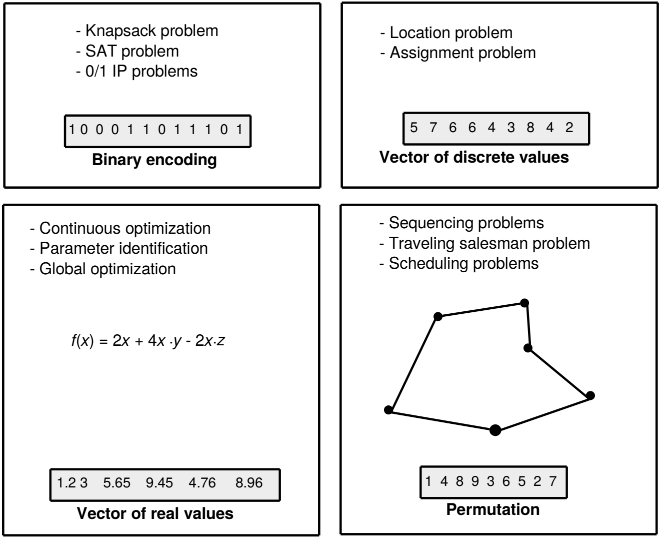

FIGURE 1.16 Some classical encodings: vector of binary values, vector of discrete values, vector of real values, and permutation.

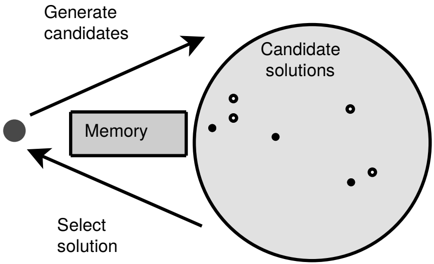

FIGURE 2.1 Main principles of single-based metaheuristics.

Algorithm 2.1 High-level template of S-metaheuristics.

Input: Initial solution s_0.
$t = 0$;
Repeat
  /* Generate candidate solutions (partial or complete neighborhood) from s_t */
  Generate($C(s_t)$) ;
  /* Select a solution from $C(s)$ to replace the current solution s_t */
  s_{t+1} = Select($C(s_t)$) ;
  $t = t + 1$;
Until Stopping criteria satisfied
Output: Best solution found.

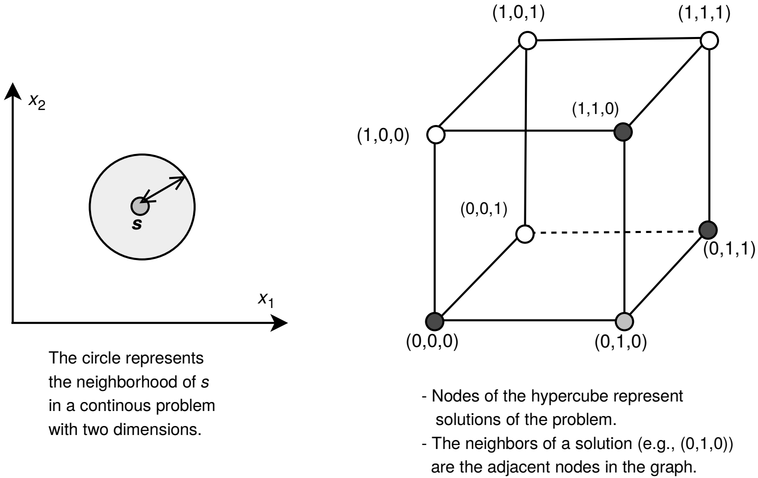

FIGURE 2.2 Neighborhoods for a continuous problem and a discrete binary problem.

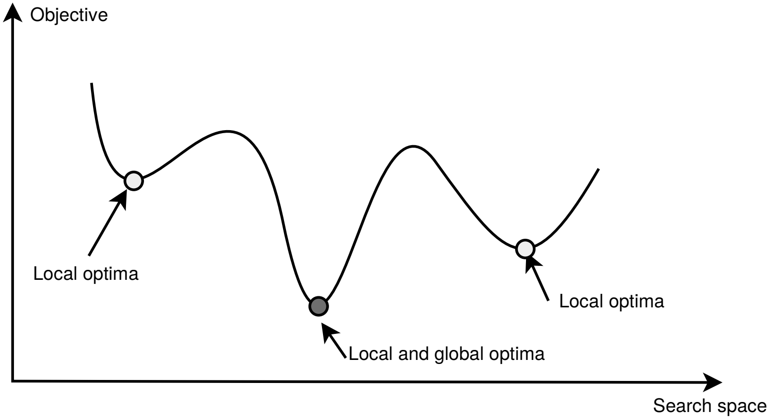

FIGURE 2.4 Local optimum and global optimum in a search space. A problem may have many global optimal solutions.

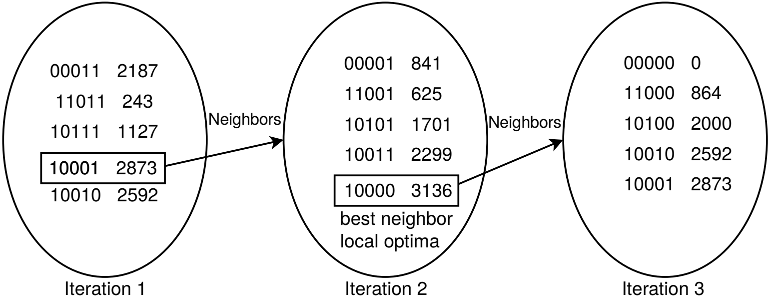

FIGURE 2.21 Local search process using a binary representation of solutions, a flip move operator, and the best neighbor selection strategy. The objective function to maximize is $x^3 - 60x^2 + 900x$. The global optimal solution is $f(01010) = f(10) = 4000$, while the final local optima found is $s = (10000)$, starting from the solution $s_0 = (10001)$.

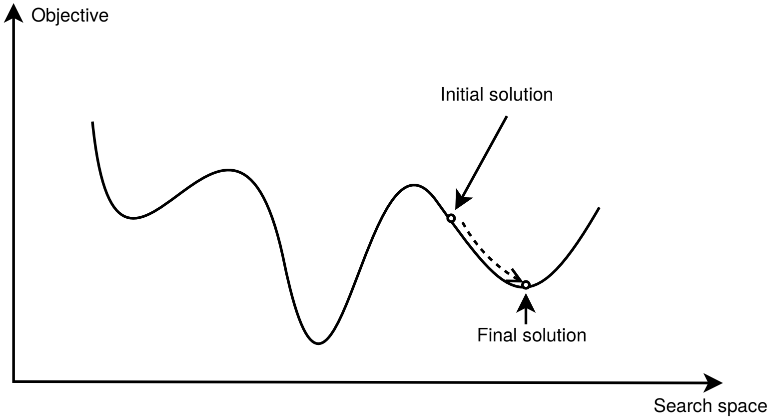

FIGURE 2.22 Local search (steepest descent) behavior in a given landscape.

Algorithm 2.2 Template of a local search algorithm.

$s = s_0$; /* Generate an initial solution s_0 */
While not Termination_Criterion **Do**
  Generate ($N(s)$) ; /* Generation of candidate neighbors */
  If there is no better neighbor **Then** Stop ;
  $s = s'$; /* Select a better neighbor $s' \in N(s)$ */
Endwhile
Output Final solution found (local optima).

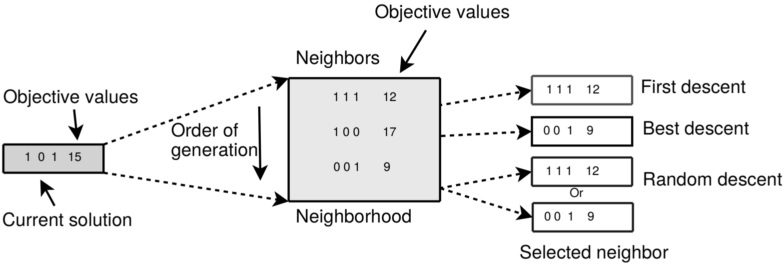

FIGURE 2.23 Selection strategies of improving neighbors.